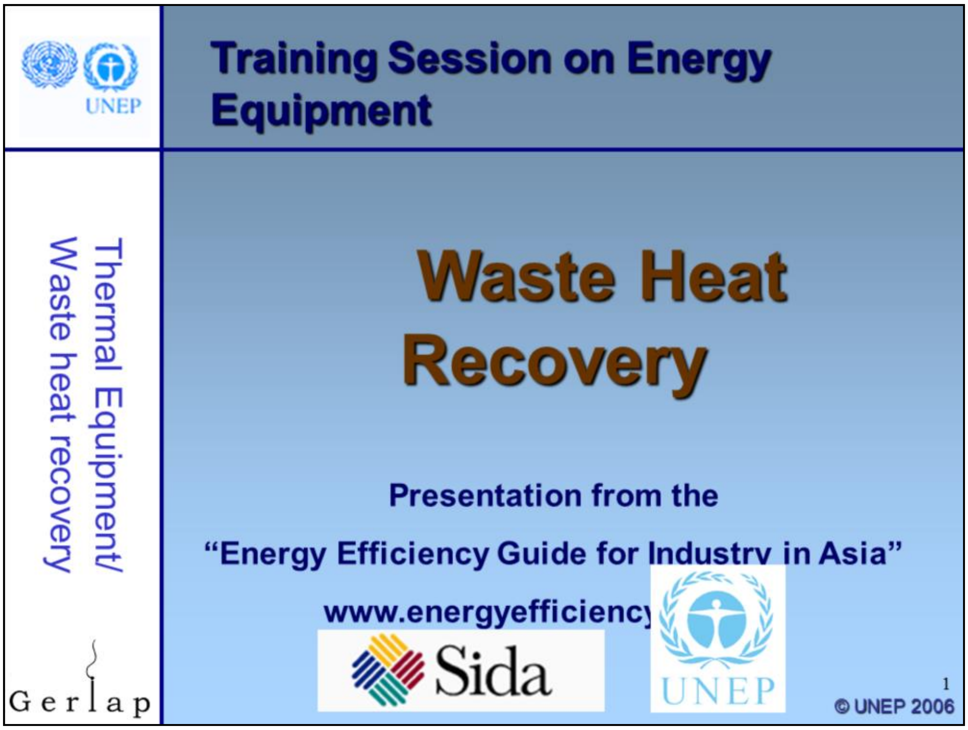


**TO THE TRAINER**

This PowerPoint presentation can be used to train people about the basics of waste heat recovery. The information on the slides is the minimum information that should be explained. The trainer notes for each slide provide more detailed information, but it is up to the trainer to decide if and how much of this information is presented also.

Additional materials that can be used for the training session are available on **www.energyefficiencyasia.org** under “Energy Equipment” and include:

•**Textbook chapter** on waste heat recovery that forms the basis of this PowerPoint presentation but has more detailed information

•**Quiz** – ten multiple choice questions that trainees can answer after the training session

•**Workshop exercise** – a practical calculation related to waste heat recovery

•**Option checklist** – a list of the most important options to maximize waste heat recovery

•**Company case studies** – participants of past courses have given the feedback that they would like to hear about options implemented at companies for each energy equipment. More than 200 examples are available from 44 companies in the cement, steel, chemicals, ceramics and pulp & paper sectors

# Training Agenda: Waste

Introduction

Type of waste heat recovery

Assessment of waste heat recovery

Thermal Equipment/
Waste heat recovery

# Introduction

Thermal Equipment/ Waste heat recovery

## What is Waste Heat?

- “Dumped” heat that can still be reused
- “Value” (quality) more important than quantity
- Waste heat recovery saves fuel

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•Waste heat is heat that has been generated in a process by fuel combustion or a chemical reaction, and then been “dumped” into the environment even though it could still be reused for some useful and economic purpose.

•The essential quality of heat is not the amount but rather its “value”. The strategy of how to recover this heat depends in part on the temperature of the waste heat gases and the economics involved.

•Large quantity of hot flue gases is generated from boilers, kilns, ovens and furnaces. If some of this waste heat could be recovered, a considerable amount of primary fuel could be saved. The energy lost in waste gases cannot be fully recovered. However, much of the heat could be recovered and loss minimized by adopting following measures as outlined in this chapter.

Thermal Equipment/ Waste heat recovery

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# Introduction

## Source and Quality

Table: Waste heat source and quality

| S. No | Source of Waste Heat | Quality of Waste Heat |
|---|---|---|
| 1 | Heat in flue gases | The higher the temperature, the greater the potential value for heat recovery |
| 2 | Heat in vapour streams | As above but when condensed, latent heat also recoverable |
| 3 | Convective & radiant heat lost from exterior of equipment | Low grade – if collected may be used for space heating or air preheats |
| 4 | Heat losses in cooling water | Low grade – useful gains if heat is exchanged with incoming fresh water |
| 5 | Heat losses in providing chilled water or in the disposal of chilled water | 1.High grade if it can be utilized to reduce demand for refrigeration<br>2.Low grade if refrigeration unit used as a form of Heat pump |
| 6 | Heat stored in products leaving the process | Quality depends upon temperature |
| 7 | Heat in gaseous & liquid | Poor if heavily contaminated & thus requiring |

Gerlap





•While considering the various potentials for heat recovery, it is useful to note all possibilities and then grade the waste heat in terms of potential value as shown in this table. There are many sources such as heat in flue gases; heat in vapour streams; convective and radiant heat that is lost from exterior of equipment; heat losses from cooling water; and losses while providing chilled water, or for that matter in the disposal of chilled water; also in heat stored in products that leave the process; and finally heat in gaseous and liquid effluents leaving the process. As you can see, the table also describes the quality of this waste heat ***(let audience reflect over the table before continuing)***

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# Introduction

Thermal Equipment/ Waste heat recovery

## High Temperature Heat Recovery

Table: Typical waste heat temperature at high temperature range from various sources

| *Types of Devices* | *Temperature (°C)* |
|---|---|
| Nickel refining furnace | 1370 – 1650 |
| Aluminium refining furnace | 650 –760 |
| Zinc refining furnace | 760 – 1100 |
| Copper refining furnace | 760 – 815 |
| Steel heating furnace | 925 – 1050 |
| Copper reverberatory furnace | 900 – 1100 |
| Open hearth furnace | 650 – 700 |
| Cement kiln (Dry process) | 620 – 730 |
| Glass melting furnace | 1000 – 1550 |
| Hydrogen plants | 650 – 1000 |
| Solid waste incinerators | 650 – 1000 |
| Fume incinerators | 650 – 1450 |

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•This table gives temperatures of waste gases from industrial process equipment in the high temperature range. All of these results from direct fuel fired processes. ***(let audience reflect over the table before continuing)***

UNEP

# Introduction

## Medium Temperature Heat Recovery

Table: Typical waste heat temperature at medium temperature range from various sources

| *Types of Devices* | *Temperature ($^0C$)* |
|---|---|
| Steam boiler exhaust | 230 – 480 |
| Gas turbine exhaust | 370 – 540 |
| Reciprocating engine exhaust | 315 – 600 |
| Reciprocating engine exhaust (turbo charged) | 230 – 370 |
| Heat treatment furnace | 425 – 650 |
| Drying & baking ovens | 230 – 600 |
| Catalytic crackers | 425 – 650 |
| Annealing furnace cooling systems | 425 – 650 |

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•This table gives the temperatures of waste gases from process equipment in the medium temperature range. Most of the waste heat in this temperature range comes from the exhaust of directly fired process units. ***(let audience reflect over the table before continuing)***

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# Introduction

Thermal Equipment/ Waste heat recovery

## Low Temperature Heat Recovery

| Source | Temperature ⁰C |
|---|---|
| Process steam condensate | 55-88 |
| Cooling water from: Furnace doors | 32-55 |
| Bearings | 32-88 |
| Welding machines | 32-88 |
| Injection molding machines | 32-88 |
| Annealing furnaces | 66-230 |
| Forming dies | 27-88 |
| Air compressors | 27-50 |
| Pumps | 27-88 |
| Internal combustion engines | 66-120 |
| Air conditioning and refrigeration condensers | 32–43 |
| Liquid still condensers | 32-88 |
| Drying, baking and curing ovens | 93-230 |
| Hot processed liquids | 32-232 |
| Hot processed solids | 93-232 |

Table: Typical waste heat temperature at low temperature range from various sources

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•This table lists some heat sources in the low temperature range. In this range it is usually not practical to extract work from the source, though steam production may not be completely excluded if there is a need for low-pressure steam. Low temperature waste heat may be useful in a supplementary way for preheating purposes. ***(let audience reflect over the table before continuing)***

## Training Agenda: Waste

Introduction

Type of waste heat recovery

Performance evaluation

Thermal Equipment/ Waste heat recovery

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Thermal Equipment/ Waste heat recovery

# Type of Waste Heat Recovery

## Commercial Waste

✓ **Recuperators**

- Heat exchange between flue gases and the air through metallic/ceramic walls
- Ducts/tubes carry combustion air for preheating
- Waste heat stream on other side

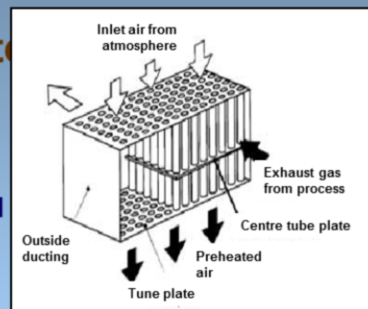



Figure 1 : Waste heat recovery using recuperator, Source: SEAV

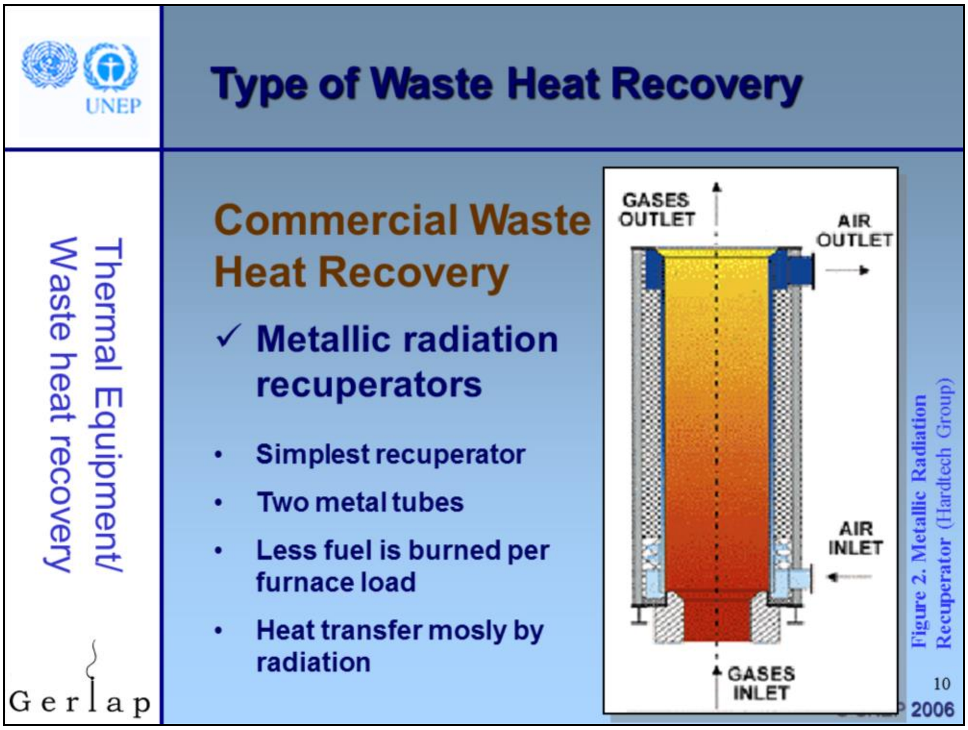




• In a recuperator, heat exchange takes place between the flue gases and the air through metallic or ceramic walls.

• Ducts or tubes carry the air for combustion to be preheated, the other side contains the waste heatstream. A recuperator for recovering

waste heat from flue gases is shown in Figure 1.




•The simplest configuration for a recuperator is the metallic radiation recuperator, which consists of two concentric lengths of metal tubing as shown in Figure 2.

•The inner tube carries the hot exhaust gases while the external annulus carries the combustion air from the atmosphere to the air inlets of the furnace burners. The hot gases are cooled by the incoming combustion air, which now carries additional energy into the combustion chamber.

•This is the energy, which does not have to be supplied by the fuel; consequently, less fuel is burned for a given furnace loading.

•The saving in fuel also means a decrease in combustion air and therefore, stack losses are decreased not only by lowering the stack gas temperatures but also by discharging smaller quantities of exhaust gas.

•The radiation recuperator gets its name from the fact that a substantial portion of the heat transfer from the hot gases to the surface of the inner tube takes place by radiative heat transfer.

•As shown in the diagram, the two gas flows are usually parallel, although the configuration would be simpler and the heat transfer would be more efficient if the flows were opposed in direction (or counterflow). The reason for the use of parallel flow is that recuperators frequently serve the additional function of cooling the duct carrying away the exhaust gases and consequently extending its service life.

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# Type of Waste Heat Recovery

Thermal Equipment/ Waste heat recovery

## Commercial Waste Heat Recovery

✓ **Convective recuperators**

- Hot gas through parallel small diameter tubes
- Tubes can be baffled to allow gas to pass over them again
- Baffling increases heat exchange but more expensive exchanger is needed

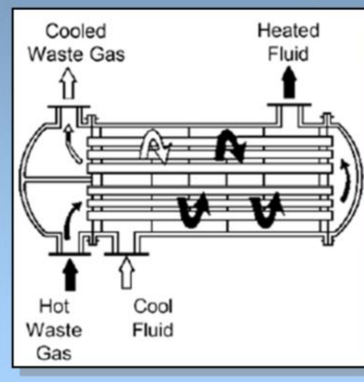


Figure 3. Convective Recuperator
(Reay, D.A., 1996)





•A second common configuration for recuperators is called the tube type or convective recuperator. As seen in this figure, the hot gases are carried through a number of parallel small diameter tubes, while the incoming air that is to be heated enters a shell surrounding the tubes and passes over the hot tubes one or more times in a direction normal to their axes.

•***(click once)*** If the tubes are baffled to allow the gas to pass over them twice, the heat exchanger is termed a two-pass recuperator. If two baffles are used, it is called a three-pass recuperator, etc.

•Baffling increases the effectiveness of heat exchange however, both the cost of the exchanger increases and the pressure drop in the combustion air path too.

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Thermal Equipment/ Waste heat recovery

# Type of Waste Heat Recovery

## Commercial Waste Heat Recovery

✓ **Radiation/convective hybrid recuperators**

- Combinations of radiation & convection
- More effective heat transfer
- More expensive but less bulky than simple metallic radiation recuperators

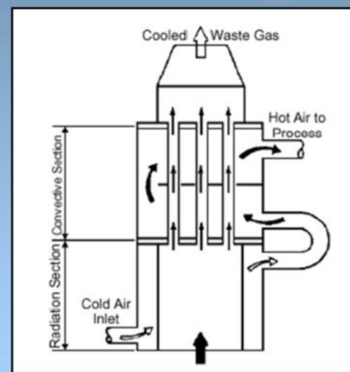


Figure 4. Hybrid Recuperator
(Reay, D.A., 1996)

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•For maximum effectiveness of heat transfer, combinations of radiation and convective designs are used, with the high-temperature radiation recuperator being first, followed by convection type.

•These are more expensive than simple metallic radiation recuperators, but are less bulky.

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# Type of Waste Heat Recovery

Thermal Equipment/
Waste heat recovery

## Commercial Waste Heat Recovery

✓ **Ceramic recuperators**

- Less temperature limitations:
  - Operation on gas side up to 1550 ◦C
  - Operation on preheated air side to 815 ◦C
- New designs
  - Last two years
  - Air preheat temperatures <700◦ C
  - Lower leakage rates

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•The principal limitation on the heat recovery of metal recuperators is the reduced life of the liner at inlet temperatures that exceed 1100◦ C. In order to overcome these temperature limitations of metal recuperators, the ceramic tube recuperators have been developed. The ceramic tube recuperators have materials that allow operation on the gas side to 1550◦ C and on the preheated air side to 815◦ C on a more or less practical basis.

•The new designs are reported to last two years with air preheat temperatures as high as 700◦ C, with lower leakage rates

Thermal Equipment/ Waste heat recovery

# Type of Waste Heat Recovery

## Regenerator

- Large capacities
- Glass and steel melting furnaces
- Time between the reversals important to reduce costs
- Heat transfer in old regenerators reduced by
  - •Dust & slagging on surfaces
  - •heat losses from the walls

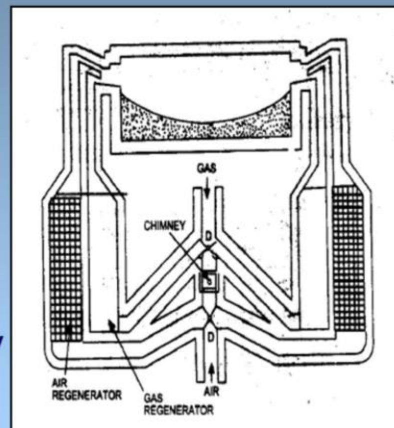


**Figure 5. Regenerator**
(Department of Coal, India, 1985)





•The regeneration that is preferred for large capacities has been very widely used in glass and steel melting furnaces.

•In a regenerator, the time between the reversals is an important aspect. Long periods would mean higher thermal storage and hence higher cost. Also, long periods of reversal result in lower average temperature of preheat and consequently reduce fuel economy.

•As the furnace becomes old, the accumulation of dust and slagging on the surfaces reduce efficiency of the heat transfer. Heat losses from the walls of the regenerator and air in leaks during the gas period and out-leaks during air period also reduces the heat transfer.

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Thermal Equipment/ Waste heat recovery

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# Type of Waste Heat Recovery

## Heat Wheels

- **Porous disk rotating between two side-by-side ducts**
- **Low to medium temperature waste heat recovery systems**
- **Heat transfer efficiency up to 85 %**

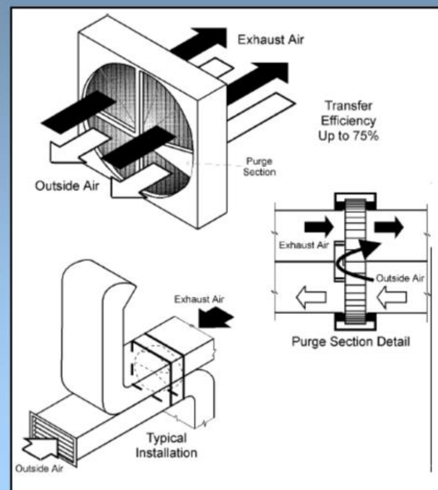


**Figure 6. Heat Wheel**
(SADC, 1999)





•A heat wheel is a sizable porous disk that is fabricated with material that has a fairly high heat capacity. The disk rotates between two side-by-side ducts: one a cold gas duct, the other a hot gas duct. The axis of the disk is located parallel to, and on the partition between, the two ducts. As the disk slowly rotates, sensible heat is transferred to the disk by the hot air and, from the disk to the cold air.

•Heat wheels are finding increasing applications in low to medium temperature waste heat recovery systems.

•The overall efficiency of sensible heat transfer for this kind of regenerator can be as high as 85 percent.

# Type of Waste Heat Recovery

Thermal Equipment/ Waste heat recovery

## Heat Pipe

- Transfer up to 100 times more thermal energy than copper
- Three elements:
  - sealed container
  - capillary wick structure
  - working fluid
- Works with evaporation and condensation

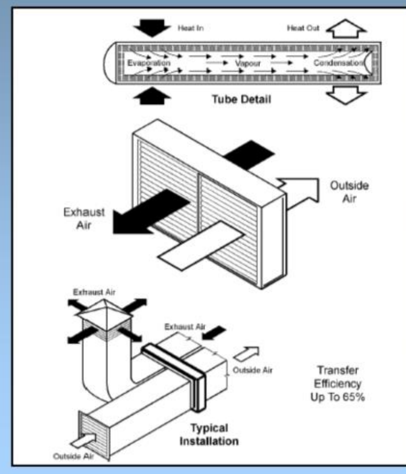


Figure 7. Heat Pipe
(SADC, 1999)




•A heat pipe can transfer up to 100 times more thermal energy than copper, the best-known conductor. In other words, heat pipe is a thermal energy absorbing and transferring system that has no moving parts and hence require minimum maintenance.

•The heat pipe comprises of three elements: a sealed container, a capillary wick structure and a working fluid. The capillary wick structure is integrally fabricated into the interior surface of the container tube and sealed under vacuum.

•Thermal energy applied to the external surface of the heat pipe is in equilibrium with its own vapour as the container tube is sealed under vacuum. Thermal energy applied to the external surface of the heat pipe causes the working fluid near the surface to evaporate instantaneously. The vapour absorbs the latent heat of vaporization and this part of the heat pipe becomes an evaporator region. The vapour then travels to the other end the pipe where the thermal energy is removed causing the vapour to condense into liquid again, thereby giving up the latent heat of the condensation. This part of the heat pipe works as the condenser region. The condensed liquid then flows back to the evaporated region.

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Thermal Equipment/ Waste heat recovery

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# Type of Waste Heat Recovery

## Heat Pipe

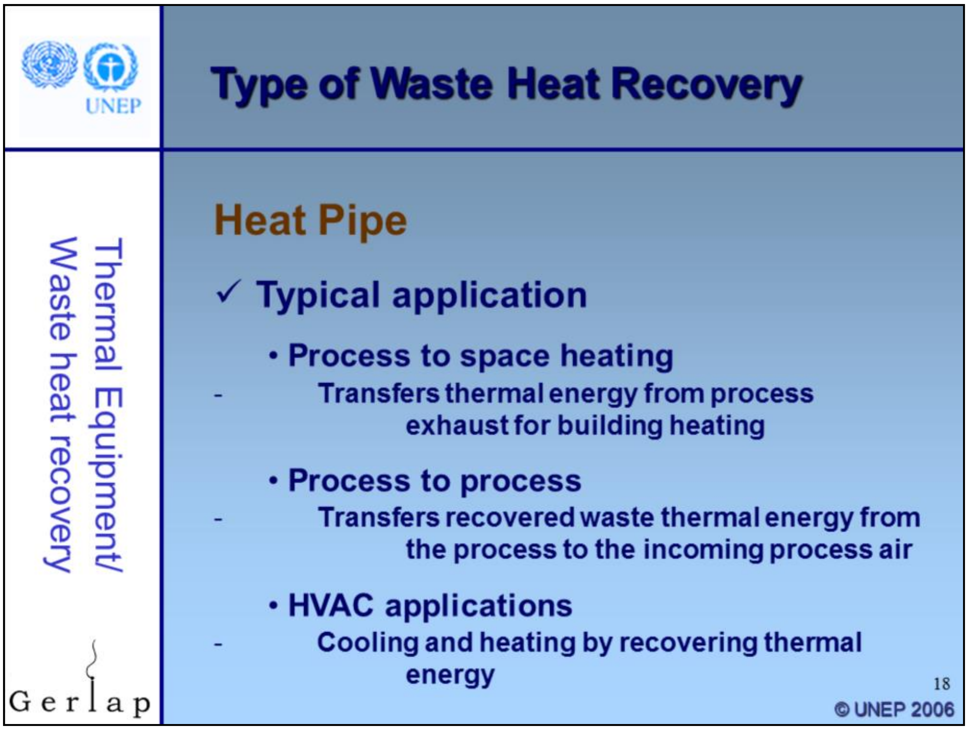
### ✓ Performance and advantage

- Lightweight and compact
- No need for mechanical maintenance, input power, cooling water and lubrication systems
- Lowers the fan horsepower requirement and increases the overall thermal efficiency of the system
- Can operate at 315 ◦C with 60% to 80% heat recovery





•The heat pipe exchanger is a lightweight compact heat recovery system. It virtually does not need mechanical maintenance, as there are no moving parts to wear out. It does not need input power for its operation and is free from cooling water and lubrication systems. It also lowers the fan horsepower requirement and increases the overall thermal efficiency of the system. The heat pipe heat recovery systems are capable of operating at 315oC. with 60% to 80% heat recovery capability.




The heat pipes are used in following industrial applications:

•First, process to space heating, where the heat pipe heat exchanger transfers the thermal energy from process exhaust for building heating. The preheated air can be blended if required. The requirement of additional heating equipment to deliver heated make up air is drastically reduced or eliminated.

•Second, process to process where the heat pipe heat exchangers recover waste thermal energy from the process exhaust and transfer this energy to the incoming process air. The incoming air thus become warm and can be used for the same process/other processes and reduces process energy consumption.

•HVAC Applications:

> - Cooling: Heat pipe heat exchangers pre-cools the building make up air in summer and thus reduces the total tones of refrigeration, apart from the operational saving of the cooling system. Thermal energy is supply recovered from the cool exhaust and transferred to the hot supply make up air.
>
> - Heating: The above process is reversed during winter to preheat the make up air.

Thermal Equipment/ Waste heat recovery

# Type of Waste Heat Recovery

## Economizer

- Utilize the flue gas heat for pre-heating the boiler feed water
- 1% fuel savings if
  - 60 °C rise of feed water
  - 200 °C rise in combustion air temp

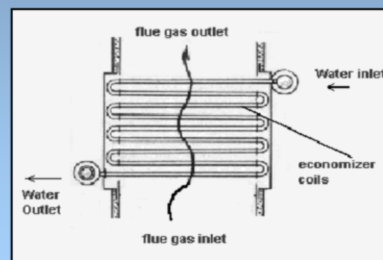



**Figure 8. Economizer**
(Bureau of Energy Efficiency, 2004)





•In the case of boiler systems, an economizer can be provided to utilize the flue gas heat for pre-heating the boiler feed water. On the other hand, in an air pre-heater, the waste heat is used to heat combustion air. In both the cases, there is a corresponding reduction in the fuel requirements of the boiler.

•For every 220 0C reduction in flue gas temperature by passing through an economizer or a pre-heater, there is 1% saving of fuel in the boiler. In other words, for every 60 0C rise in feed water temperature through an economizer, or 200 0C rise in combustion air temperature through an air pre-heater, there is 1% saving of fuel in the boiler.

# Type of Waste Heat Recovery

## Economizer

✓ **Shell and tube heat exchanger**

- Used when the medium containing waste heat is a liquid or a vapor that heats another liquid
- Shell contains the tube bundle, and usually internal baffles to direct the fluid
- Vapor contained within the shell

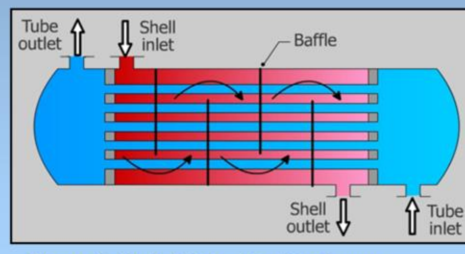



Figure 9. Shell & Tube Heat Exchanger
(King Fahad University of Petroleum & Minerals, 2003)





•When the medium that contains waste heat is a liquid or a vapor that heats another liquid, then the shell and tube heat exchanger must be used since both paths must be sealed to contain the pressures of their respective fluids.

•The shell contains the tube bundle and usually internal baffles to direct the fluid in the shell over the tubes in multiple passes. The shell is inherently weaker than the tubes so that the higher-pressure fluid is circulated in the tubes while the lower pressure fluid flows through the shell.

•***(click once)*** When a vapor contains the waste heat, it usually condenses and thereby gives up its latent heat to the liquid being heated. In this application, the vapor is almost invariably contained within the shell. If the reverse is attempted, the condensation of vapors within small diameter parallel tubes causes flow instabilities. Tube and shell heat exchangers are available in a wide range of standard sizes with many combinations of materials for the tubes and shells.

# Type of Waste Heat Recovery

Thermal Equipment/ Waste heat recovery

## Plate Heat Exchanger

- Parallel plates forming a thin flow pass
- Avoids high cost of heat exchange surfaces
- Corrugated plates to improve heat transfer
- When directions of hot and cold fluids are opposite, the arrangement is counter current

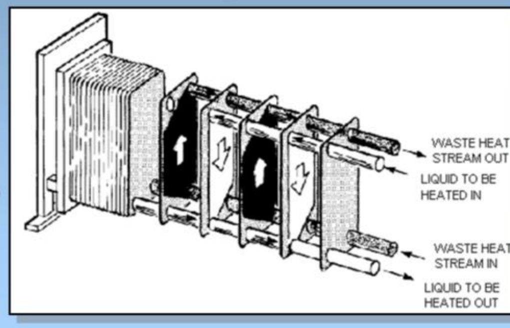



Figure 10. Plate Heat Exchanger
(Canada Agriculture and Agri-Food)





•The cost of heat exchange surfaces is a major cost factor when the temperature differences are not large. One way of meeting this problem is the plate type heat exchanger. This consists of a series of separate parallel plates forming thin flow pass. Each plate is separated from the next by gaskets. The hot stream passes in parallel through alternative plates whilst the liquid to be heated passes in parallel between the hot plates.

•***(click once)*** To improve heat transfer the plates are corrugated.

•Hot liquid passing through a bottom port in the head is permitted to pass upwards between every second plate while cold liquid at the top of the head is permitted to pass downwards between the odd plates. When the directions of hot and cold fluids are opposite, the arrangement is described as counter current.

•Typical industrial applications for the plate heat exchanger are pasteurisation section in milk packaging plant and evaporation plants in food industry.

# Type of Waste Heat Recovery

## Plate Heat Exchanger

✓ Run around coil exchanger

- Heat transfer from hot to colder fluid via *heat transfer fluid*
- One coil in hot stream
- One coil in cold stream

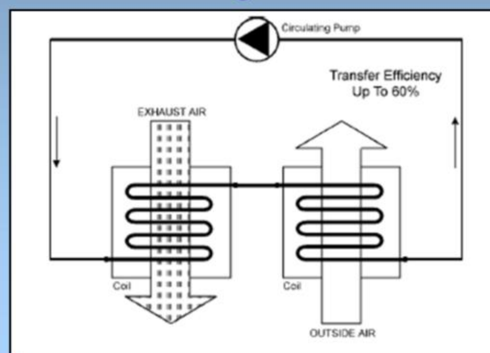



Figure 11. Run Around Coil Exchanger (SADC , 1999)





•The run around coil exchanger is quite similar in principle to the heat pipe exchanger. The heat from hot fluid is transferred to the colder fluid via an intermediate fluid known as the "heat transfer fluid".

•One coil of this closed loop is installed in the hot stream while the other is in the cold stream. Circulation of this fluid is maintained by means of la circulating pump.

•Typical industrial applications are heat recovery from ventilation, air conditioning and low temperature heat recovery.

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Thermal Equipment/ Waste heat recovery

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# Type of Waste Heat Recovery

## Plate Heat Exchanger

### ✓ Waste heat boiler

- Water tube boiler: hot exhaust gases pass over parallel tubes with water
- Capacities: 25 m3 to 30,000 m3 /min of exhaust gas

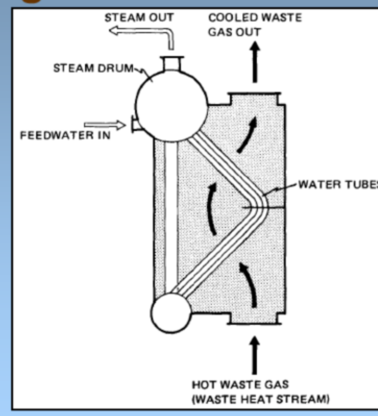



**Figure 12. Two-Pass Water Tube Waste Heat Recovery Boiler**
(Canada Agriculture and Agri-Food)

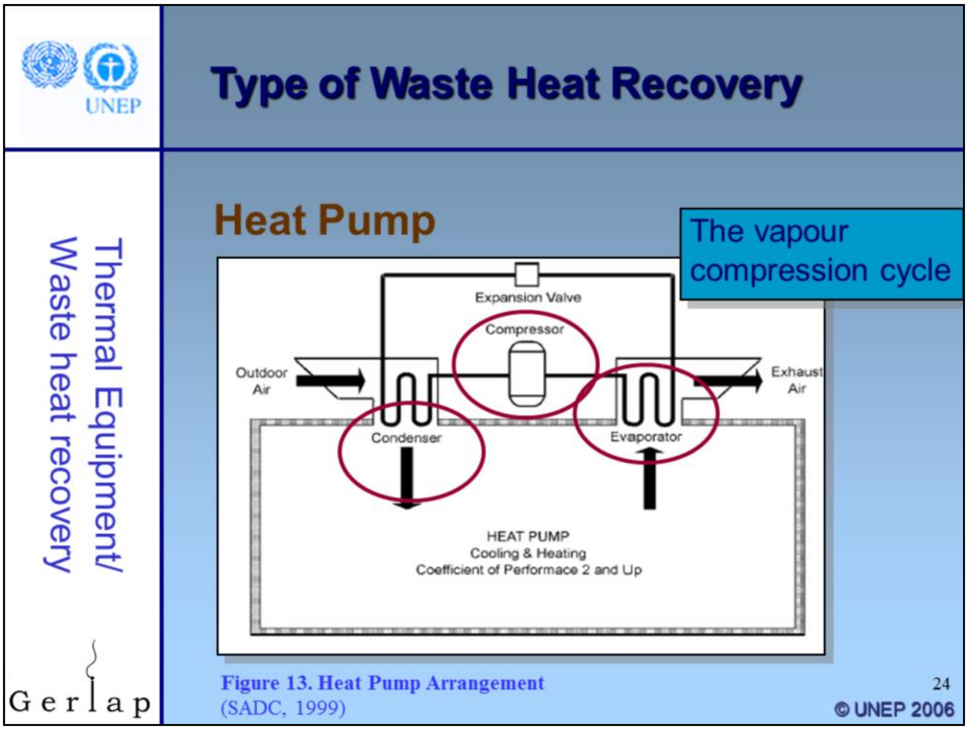




•Waste heat boilers are ordinarily water tube boilers in which the hot exhaust gases from gas turbines, incinerators, etc pass over a number of parallel tubes that contain water. The water is vaporized in the tubes and collected in a steam drum from which it is drawn off for use as heating or processing steam.

•This figure illustrates a mud drum and a steam drum. A mud drum is a set of tubes over which the hot gases make a double pass. A steam drum collects the steam generated above the water surface. The pressure at which the steam is generated and the rate of steam production depends on the temperature of waste heat. The pressure of a pure vapor in the presence of its liquid is a function of the temperature of the liquid from which it is evaporated.

•Waste heat boilers are built in capacities from 25 m3 almost 30,000 m3 /min. of exhaust gas.




- Heat must flow spontaneously "downhill". This means that the heat must flow from a system at high temperature to one at a lower temperature. However, when the energy is repeatedly transferred or transformed, it becomes less and less available for use. Eventually the energy has such low intensity that it is no longer available at all to perform a useful function.
- ***(Click once)*** It is possible to reverse the direction of spontaneous energy flow by the use of a thermodynamic system known as a heat pump.
- The majority of heat pumps work on the principle of the vapour compression cycle ***(click once).*** In this cycle, the circulating substance is physically separated from the source and is re-used in a cyclical fashion, therefore called 'closed cycle'.
- We will run through the various processes that take place in the heat pump. First, in the evaporator ***(click once and circle will appear)*** the heat is extracted from the heat source to boil the circulating substance. The circulating substance is compressed by the compressor ***(click once and circle will appear)*** that raises its pressure and temperature. The low temperature vapor is compressed by a compressor, which requires external work. The work done on the vapor raises its pressure and temperature to a level where its energy becomes available for use

- The heat is delivered to the condenser ***(click once and circle will appear)***. The pressure of the circulating working fluid is reduced back to the evaporator condition in the throttling valve where the cycle repeats.

# Type of Waste Heat Recovery

## Heat Pump

- ✓ Developed as a space heating system
- ✓ Can upgrade heat >2X the energy consumed by the device
- ✓ Most promising when heating and cooling capabilities are combined




•The heat pump was developed as a space heating system where low temperature energy from the ambient air, water, or earth is raised to heating system temperatures by doing compression work with an electric motor-driven compressor.

•The heat pump has the ability to upgrade heat to a value more than twice that of the energy consumed by the device. The potential for application of heat pump is growing and number of industries have been benefited by recovering low grade waste heat by upgrading it and using it in the main process stream.

•Heat pump applications are most promising when both the heating and cooling capabilities can be used in combination. One example of this is a plastics factory where chilled water from a heat is used to cool injection-moulding machines whilst the heat output from the heat pump is used to provide factory or office heating. Other examples include product drying, maintaining dry atmosphere for storage and drying compressed air.

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Thermal Equipment/ Waste heat recovery

# Type of Waste Heat Recovery

## Heat Pump

✓ **Thermo compressor**

- Compress low-pressure steam by very high-pressure steam and reuse as medium pressure steam
- Nozzle for acceleration of HP steam to a high velocity fluid.

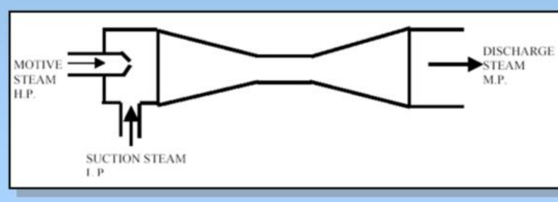



Figure: Thermo compressor





•Often, very low pressure steam are reused as water after condensation due to the lack of any better option of reuse. In many cases it is feasible to compress this low-pressure steam by very high-pressure steam and reuse it as a medium pressure steam. The major energy in steam is in its latent heat value and thus thermocompressing would give a large improvement in waste heat recovery.

•The thermocompressor is a simple equipment with a nozzle where HP steam is accelerated into a high velocity fluid. This entrains the LP steam by momentum transfer and then recompresses in a divergent venturi. Thermo compressors are typically used in evaporators where the boiling steam is recompressed and used as heating steam.

# Training Agenda: Waste

Introduction

Type of waste heat recovery

Assessment of waste heat recovery

Thermal Equipment/ Waste heat recovery

# Assessment of waste heat recovery

## Heat Losses

**Quality:**

- Higher temperatures = Higher quality = Lower heat recovery costs

**Quantity:**

- The amount of recoverable heat can be calculated as:

$$Q = V \times \rho \times Cp \times \Delta T$$

Q = heat content in kCal
V = the flow rate of the substance in m3/hr
ρ = density of the flue gas in kg/m3
Cp = the specific heat of the substance in kCal/kg oC
ΔT = the temperature difference in oC




•Heat losses can be divided into qualitative and quantitative.

•Starting with qualitative, depending on the type of process, waste heat can be rejected at virtually any temperature. From that of chilled cooling water to high temperature waste gases from an industrial furnace or kiln. Usually, the higher the temperature, the higher the quality and more cost effective is the heat recovery.

•***(Click once)*** In any heat recovery situation it is essential to know the amount of heat that is recoverable and also how it can be used. Heat recovery from heat treatment furnace can be calculated as Q = V x ρ x Cp x ΔT. Where, Q is the heat content; V is the flowrate of the substance; ρ is density of the flue gas; Cp is the specific heat of the substance; ΔT is the temperature difference; and Cp is 0.24 kCal/kg/oC

# Assessment of waste heat recovery

Thermal Equipment/ Waste heat recovery

## Heat Saving Calculation Example

**Saving money by recovering heat from hot waste water:**

- Discharge of the waste water is 10000 kg/hr at 75◦C
- Preheat 10000 kg/hr of cold inlet water of 20◦C
- A heat recovery factor of 58%
- An operation of 5000 hours per year

**The annual heat saving (Q) is:**

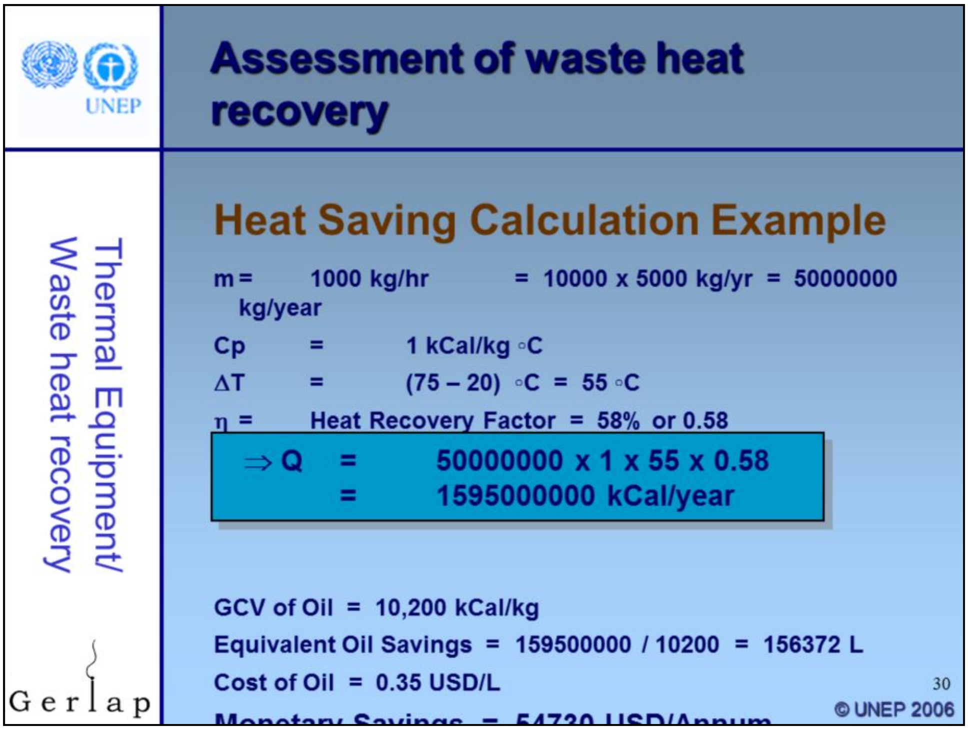
$$Q = m \times Cp \times \Delta T \times \eta$$




•We will go through a heat saving calculation example.

•A large paper manufacturing company identifies an opportunity to save money by recovering heat from hot waste water. The discharge of the waste water from the operation ranges is 10000 kg/hr at 75 ◦ C. Rather than discharging this water to drain, it was decided to preheat the 10000 kg/hr of cold inlet water having a yearly average temperature of 20 ◦ C, by passing it through a counterflow heat exchanger with automatic back flushing to reduce fouling. Based on a heat recovery factor of 58% and an operation of 5000 hours per year.

•***(Click once)*** The annual heat saving known as Q can be calculated as: $Q = m \times Cp \times \Delta T \times \eta$




•m = 1000 kg/hr = 10000 x 5000 kg/yr = 50000000 kg/year.

•Cp = 1 kCal/kg◦ C.

•ΔT = (75 – 20) 0C = 55◦ C.

•η = Heat Recovery Factor = 58% or 0.58

•***(click once)*** This gives us the annual heat saving Q as 50000000 x 1 x 55 x 0.58 which equals 1595000000 kCal/year

•***(click once)*** Furthermore, this means: GCV of oil = 10,200 kCal/kg. Equivalent oil savings = 159500000 / 10200 = 156372 L. Cost of oil = 0.35 USD/L.

•The monetary savings would be 54730 USD/Annum


Thank you for your attention!

Thermal Equipment/ Waste heat recovery

## Disclaimer and References

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